

2E Phase 4 Documentation

Status: At the moment, we have common subexpression elimination implemented. Our CSE optimization does not operate on global variables as it would require an overhaul of our current code base, and we believe time would be better spent getting other optimizations to work first. Additionally, we have algebraic simplification implemented as well, even though this is not a dataflow optimization. Our plan after this is to try and generalize the CSE implementation to a lattice structure, which should ease the implementation of the other dataflow optimizations, then pursue optimizations for phase 5.

Questions/Concerns: We do not have any questions or concerns for the TAs to address.

Team Dynamics: Noah was responsible for implementing algebraic simplification. Matt implemented CSE at the block level, and Noah expanded upon this implementation to enable cross-block CSE. Both Noah and Matt debugged the implementation after expansion, with some insight provided by Julius and Adriano.